

Just lg

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```
# libraries/etc
library(tidymodels)
```

```
## -- Attaching packages ----- tidymodels 1.2.0 --
```

```
## v broom      1.0.7    v recipes      1.1.0
## v dials      1.3.0    v rsample      1.2.1
## v dplyr      1.1.4    v tibble       3.2.1
## v ggplot2    3.5.1    v tidyr        1.3.1
## v infer      1.0.7    v tune         1.2.1
## v modeldata  1.4.0    v workflows    1.1.4
## v parsnip    1.2.1    v workflowsets 1.1.0
## v purrr      1.0.2    v yardstick    1.3.1
```

```
## -- Conflicts ----- tidymodels_conflicts() --
```

```
## x purrr::discard() masks scales::discard()
## x dplyr::filter()  masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## x recipes::step() masks stats::step()
## * Dig deeper into tidy modeling with R at https://www.tmr.org
```

```
library(ggplot2)
library(vip)
```

```
## Warning: package 'vip' was built under R version 4.4.2
```

```
##
```

```
## Attaching package: 'vip'
```

```
## The following object is masked from 'package:utils':
```

```
##
```

```
##      vi
```

```
options(scipen=4)
```

```
## Data Processing
```

```
# Initial data import
```

```
customer_data <- read.csv("~/00 Merrimack/DSE6111/Course Project/customer_data_1.csv", header = TRUE)
```

```
customer_data$Attrition_Flag <- ifelse(customer_data$Attrition_Flag == "Attrited Customer", 1, 0)
customer_data$Attrition_Flag <- factor(customer_data$Attrition_Flag)
```

```
# check levels
levels(customer_data$Attrition_Flag)
```

```
## [1] "0" "1"
```

```
# set seed
set.seed(10)
```

```
# get rid of ClientNum identifier
customer_data <- customer_data[ -c(1) ]
```

```
# create data split
data_split <- initial_split(customer_data, strata = "Attrition_Flag", prop = 0.75)
```

```
# create training and test sets
training_set <- training(data_split)
test_set <- testing(data_split)
```

```
# row count per set
nrow(training_set)
```

```
## [1] 7595
```

```
nrow(test_set)
```

```
## [1] 2532
```

```
# use cv folds and tune
cv_set <- vfold_cv(training_set, v = 10)
```

```
# create recipe for preprocessing
attrition_recipe_lg <-
  recipe(
    Attrition_Flag ~ .,
    data = training_set,
  ) %>%
  step_dummy(all_nominal_predictors()) %>%
  step_center(all_predictors()) %>%
  step_scale(all_predictors())
```

```
# compare to the null model
null_classification <- null_model() %>%
  set_engine("parsnip") %>%
  set_mode("classification")

null_lg <- workflow() %>%
  add_recipe(attrition_recipe_lg) %>%
```

```

add_model(null_classification) %>%
fit_resamples(
  cv_set
)

null_metrics <- null_lg %>% collect_metrics()

null_lg %>% show_best(metric = "roc_auc")

```

```

## # A tibble: 1 x 6
##   .metric .estimator mean      n std_err .config
##   <chr>   <chr>      <dbl> <int>   <dbl> <chr>
## 1 roc_auc binary      0.5     10      0 Preprocessor1_Model1

```

```

null_lg %>% show_best(metric = "accuracy")

```

```

## # A tibble: 1 x 6
##   .metric .estimator mean      n std_err .config
##   <chr>   <chr>      <dbl> <int>   <dbl> <chr>
## 1 accuracy binary    0.839     10 0.00369 Preprocessor1_Model1

```

```

null_metrics

```

```

## # A tibble: 3 x 6
##   .metric .estimator mean      n std_err .config
##   <chr>   <chr>      <dbl> <int>   <dbl> <chr>
## 1 accuracy binary    0.839     10 0.00369 Preprocessor1_Model1
## 2 brier_class binary    0.135     10 0.00250 Preprocessor1_Model1
## 3 roc_auc binary     0.5      10 0      Preprocessor1_Model1

```

```

““

```